



Sydney Girls High School

2009

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics

Extension 1

This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 2009 HSC Examination Paper in this subject.

--	--	--	--	--	--	--	--	--	--

Candidate Number

General Instructions

- Reading Time - 5 minutes
- Working time - 2 hours
- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Diagrams are not to scale
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

Question 1 (12 marks)

- (a) The remainder when the polynomial x^4 is divided by $x + a$ is 16.
Find the value of a

Marks

2

- (b) Differentiate $\sin^{-1}\left(\frac{x}{2}\right)$ with respect to x

1

- (c) Evaluate $\int_0^3 \frac{1}{x^2 + 9} dx$

3

- (d) The interval AB, where A is (2,1) and B is (3,2) is divided internally in the ratio 4:3 by the point P(x,y). Find the values of x and y .

2

- (e) Evaluate $\int_0^1 xe^{x^2} dx$ Leave your answer in exact form.

2

- (f) Let $f(x) = \sqrt{x^2 - 5x + 6}$ What is the domain of $f(x)$?

2

Question 2 (12 marks)

(a) Use the substitution $u = 1 - \sin x$ to evaluate $\int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \cos x \sqrt{1 - \sin x} \, dx$ 3

(b) A particle moves on the x -axis with velocity v .
Velocity is given by $v = x - 3$
Using the fact that $a = \frac{d}{dx}(\frac{1}{2}v^2)$, find the acceleration of the particle at $x = 4$ 3

(c) The polynomial $P(x) = x^2 + ax + b$ has a zero at $x = 2$. When $P(x)$ is divided by $x - 1$, the remainder is 2. Find the value of a and b 3

(d) The function $f(x) = \sin x + \log x$ has a zero near $x = 0.5$
Use one application of Newton's method to obtain another approximation to this zero. Give your answer correct to three decimal places. 3

Question 3 (12 marks)

(a) (i) Sketch the graph of $y = x(x-2)(x-3)$ showing the x and y intercepts. Do not use calculus. 2

(ii) Hence, or otherwise, solve $x^3 + 6x \geq 5x^2$ 2

(b) Use mathematical induction to prove that, for integers $n \geq 1$
 $1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$ 3

(c) The radius of a spherical balloon is expanding at a constant rate of 6 cm s^{-1}
($V = \frac{4}{3}\pi r^3, S = 4\pi r^2$)

(i) At what rate is the volume of the balloon expanding when its radius is 4cm? 2

(ii) At what rate is the surface area of the balloon expanding if the rate of change of volume is $15 \text{ cm}^3 \text{ s}^{-1}$ 3

Marks

Question 4 (12 marks)

- (a) A freshly baked chocolate cake is cooling in a room of constant temperature of 20°C . At time t minutes its temperature T decreases according to the equation $\frac{dT}{dt} = -k(T - 20)$ where k is a positive constant. The initial temperature of the cake is 150°C and it cools to 100°C after 15 minutes.

(i) Verify that $T = 20 + Ae^{-kt}$ is a solution of this equation, where A is a constant 1

(ii) Find the values of A and k , giving k correct to three decimal places. 2

(iii) How long will it take for the temperature of the cake to cool to 25°C ? Give your answer to the nearest minute. 2

- (b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$

(i) Show that the gradient of PQ is $\frac{p+q}{2}$ 1

(ii) Show that if PQ passes through the focus then $pq = -1$ 2

(iii) Find the equation of the locus of the midpoint of PQ if PQ is a focal chord. 2

(c) Solve $\frac{5}{x-1} > 2$ 2

Marks

Question 5 (12 marks)

(a) Let $f(x) = x^2 - 2x$ for $x \geq 1$. This function has an inverse, $f^{-1}(x)$ 3

(i) Sketch the graphs of $y = f(x)$, $y = f^{-1}(x)$ and $y = x$ on the same set of axes. (Use the same scale on both axes)

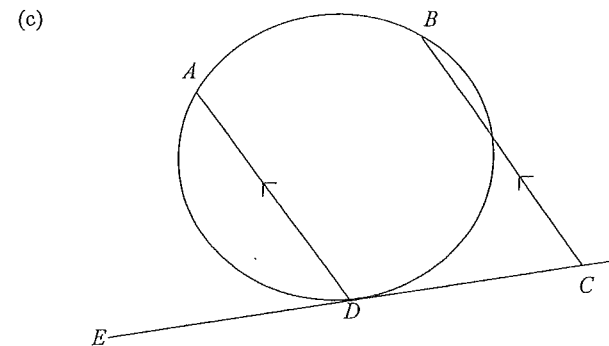
(ii) Find an expression for $f^{-1}(x)$ 2

(iii) Evaluate $f^{-1}(2)$ 1

(b) A particle is moving in simple harmonic motion in a straight line. 3

Its amplitude is 3m and its period is $\frac{\pi}{2}$ seconds.

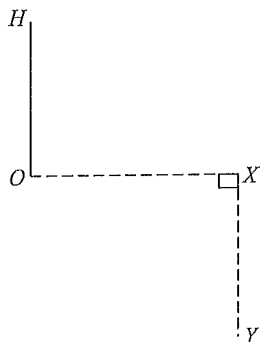
Find the maximum speed and maximum acceleration of the motion.



In the diagram above, $AD \parallel BC$ and the line EC is a tangent to the circle at D . Copy or trace the diagram. Prove that $BD^2 = AD \cdot BC$ 3

Question 6 (12 marks)

- (a) From a point X due east of a tower, the angle of elevation of the top of the tower H is 31° . From another point Y due south of X , the angle of elevation of H is 25° . The distance XY is 150m.



- (i) Copy or trace the diagram, adding the given information to your diagram
- (ii) Hence find the height of the tower

- (b) Find all values of θ in the range $0 \leq \theta \leq 2\pi$ for which $\sqrt{3} \cos \theta - \sin \theta = 1$

- (c) Find the equation of the normal to the curve $y = \frac{8}{x^2 + 4}$ at the point $(2, 1)$

- (ii) Show that the normal in (i) does not touch the curve again.

Marks

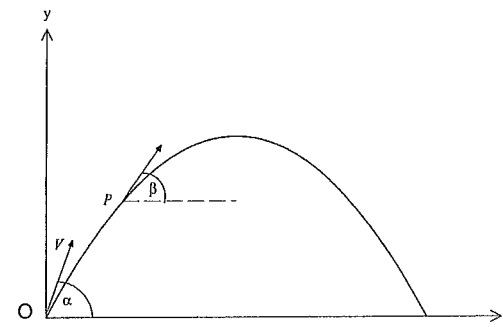
Question 7 (12 marks)

- (a) $y = f(x)$ is an odd function
 $f(x) \geq 0$ if $x \geq 0$.

$$\int_0^a f(x) dx = a^2 \times 2^a \text{ if } a > 0$$

Find the area bounded by $y = f(x + 3)$, $x = -4$, $x = 4$ and the x axis

- (b) A particle is projected from a point O on horizontal ground, with speed $V \text{ ms}^{-1}$ at an angle of elevation to the horizontal of α



Its equations of motion are $\ddot{x} = 0$, $\ddot{y} = -g$

- (i) Show that $x = Vt \cos \alpha$ and $y = Vt \sin \alpha - \frac{1}{2}gt^2$

- (ii) Show that the time of flight of the particle is $\frac{2V \sin \alpha}{g}$

- (iii) The particle reaches a point P , as shown, where the direction of the flight makes an angle β with the horizontal.

Show that the time taken to travel from O to P is $\frac{V \sin(\alpha - \beta)}{g \cos \beta}$ seconds

- (iv) Consider the case where $\beta = \frac{\alpha}{2}$. If the time taken to travel from O to P is then one-third of the total time of flight, find the value of α

1

3

4

2

2

Marks

3

2

2

2

3

- End of Exam -

Ext ①

Question 1

a) $f(x) = x^4$

divided by $x+4$

$R = 16$

$f(-a) = (-a)^4$

$= a^4$

$a^4 = 16$

$\therefore a = \pm 2$

②

f) Domain:

$x^2 - 5x + 6 > 0$

$(x-3)(x-2) > 0$

$x < 2$ and

$x > 3$ ②



b) $\frac{d}{dx} \left(\sin^{-1} \frac{x}{2} \right) = \frac{1}{\sqrt{4-x^2}}$ ①

c) $\int_0^3 \frac{1}{x^2+9} dx = \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^3$

$= \frac{1}{3} \tan^{-1} 1 - \frac{1}{3} \tan^{-1} 0$

$= \frac{1}{3} \cdot \frac{\pi}{4} = 0$

$= \frac{\pi}{12}$

③

d) $x = \frac{nx_1 + mx_2}{m+n}$ $y = \frac{ny_1 + my_2}{m+n}$ $m:n = 4:3$

$x = \left(\frac{3(2) + 4(3)}{7} \right)$ $y = \left(\frac{3(1) + 4(2)}{7} \right)$

$x = \left(\frac{6+12}{7} \right)$ $y = \left(\frac{3+8}{7} \right)$

$x = \frac{18}{7}$ $y = \frac{11}{7}$

$P\left(\frac{18}{7}, \frac{11}{7}\right)$

②

3U Q2

(a) $u = 1 - \sin x$

$\frac{du}{dx} = -\cos x$

$-du = \cos x dx$

$\int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \cos x \sqrt{1 - \sin x} dx$

$= \int_0^{\frac{1}{2}} -\sqrt{u} du$

$= -\left[\frac{2u^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^{\frac{1}{2}}$

$= -\left(\frac{2}{3} \left(\frac{1}{2} \right)^{\frac{3}{2}} - 0 \right)$

$= -\left(\frac{\sqrt{1}}{2} \right)^3 = -\left(\frac{1}{\sqrt{2}} \right)^3 = -\frac{1}{2\sqrt{2}}$

$= -\frac{2}{3} \left(\frac{1}{2\sqrt{2}} \right)$

$= -\frac{1}{3\sqrt{2}}$

✓

(d) $P(x) = \sin x + \log x$

$P(0.5) = \sin 0.5 + \log 0.5$

$P'(x) = \cos x + \frac{1}{x}$

$P'(0.5) = \cos 0.5 + \frac{1}{0.5}$

$= \cos 0.5 + 2$

$x = a - \frac{P(a)}{P'(a)}$

$= 0.5 - \frac{\sin 0.5 + \log 0.5}{\cos 0.5 + 2}$

≈ 0.574

✓

when $x = \frac{\pi}{6}$
 $u = 1 - \sin \frac{\pi}{6}$
 $= 1 - \frac{1}{2} = \frac{1}{2}$
 when $x = \frac{\pi}{2}$
 $u = 1 - \sin \frac{\pi}{2}$
 $= 1 - 1 = 0$

(b) $V = x-3$

$V^2 = (x-3)^2$

$\frac{1}{2} V^2 = \frac{1}{2} (x^2 - 6x + 9)$

$= \frac{1}{2} x^2 - 3x + \frac{9}{2}$

$a = \frac{d}{dx} \left(\frac{1}{2} V^2 \right)$

$= x-3$

when $x=4$

$a = 4-3$

$= 1$

✓

(c) $P(x) = x^2 + ax + b$

Factor $(x-2)$

$P(2) = 4 + 2a + b$

$0 = 4 + 2a + b$

$\therefore b = -2a - 4$ (1)

$P(1) = 1 + a + b$

$2 = 1 + a + b$

$b = -a + 1$ (2)

(1) = (2):

$-2a - 4 = -a + 1$

$-5 = a$

Sub a in (2):

$b = -(-5) + 1$

$= 6$

$\therefore a = -5, b = 6$

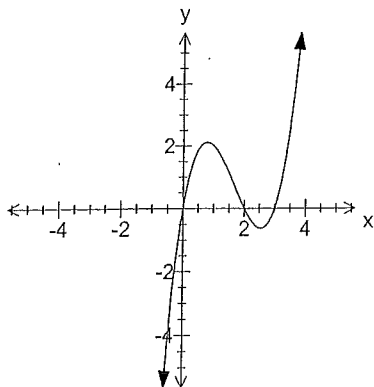
✓

Extension 1 – Trial 2009

Question 3 Solutions

a.

i.



ii.

From part i. $0 \leq x \leq 2$, or $x \geq 3$

b.

Step 1 – Prove true for $n=1$

$$\begin{aligned} LHS &= 1 \times 2 \\ &= 2 \end{aligned}$$

$$RHS = \frac{1(1+1)(1+2)}{3}$$

$$= \frac{6}{3}$$

$$= 2$$

$$= LHS$$

$$\therefore \text{true for } n=1$$

Step 2 – Assume true for $n=k$

$$1 \times 2 + 2 \times 3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

Step 3 – Prove true for $n=k+1$

Required to prove that:

$$1 \times 2 + 2 \times 3 + \dots + k(k+1) + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$$

$$LHS = 1 \times 2 + 2 \times 3 + \dots + k(k+1) + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3}$$

$$= \frac{(k+3)(k+1)(k+2)}{3}$$

$$= RHS$$

$$\therefore \text{true for } n=k+1$$

Therefore the statement is true for all positive integral values of n by induction.

c.

$$\frac{dr}{dt} = 6$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dS}{dr} = 8\pi r$$

i.

$$\begin{aligned} \frac{dV}{dt} &= \frac{dr}{dt} \times \frac{dV}{dr} \\ &= 6 \times 4\pi r^2 \\ &= 24\pi r^2 \end{aligned}$$

when $r = 4$

$$\begin{aligned} \frac{dV}{dt} &= 24 \times \pi \times 4^2 \\ &= 384\pi \text{ cm}^3 \text{ s}^{-1} \end{aligned}$$

ii.

$$\frac{dV}{dt} = 15$$

$$15 = 24\pi r^2$$

$$r^2 = \frac{15}{24\pi}$$

$$r = \sqrt{\frac{15}{24\pi}}$$

$$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$$

$$= 8\pi r \times 6$$

$$= 48\pi r$$

$$= 48\pi \times \sqrt{\frac{15}{24\pi}}$$

$$= 67.26 \text{ cm}^2 \text{ s}^{-1}$$

Question 4

(a). (i). $T = 20 + Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(20 + Ae^{-kt} - 20)$$

$$= -k(T - 20)$$

(ii).

$$T = 20 + Ae^{-kt}$$

$$150 = 20 + A$$

$$\therefore A = 130$$

$$T = 20 + 130e^{-kt}$$

$$100 = 20 + 130e^{-15k}$$

$$e^{-15k} = \frac{8}{13}$$

$$k = -\frac{1}{15} \ln\left(\frac{8}{13}\right)$$

$$\therefore k \approx 0.032$$

(iii).

$$T = 25 \text{ when } t = ?$$

$$T = 20 + 130e^{-0.032t}$$

$$25 = 20 + 130e^{-0.032t}$$

$$e^{-0.032t} = \frac{5}{130}$$

$$t = \ln\left(\frac{5}{130}\right) \div (-0.032)$$

$$\therefore t \approx 101.8 \text{ (i.e. 102 minutes)}$$

(b). (i). $m_{PQ} = \frac{ap^3 - aq^3}{2ap - 2aq}$

$$= \frac{a(p+q)(p-q)}{2a(p-q)}$$

$$= \frac{p+q}{2}$$

(ii).

Equation of line PQ:

$$y - ap^3 = \frac{p+q}{2}(x - 2ap)$$

If PQ passes through the focus $(0, a)$ then:

$$a - ap^3 = \frac{p+q}{2}(-2ap)$$

$$2a - 2ap^3 = -2ap^2 - 2apq$$

$$2a = -2apq$$

$$pq = -1$$

(iii).

Midpoint of PQ is:

$$M = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$= \left(a(p+q), \frac{a}{2}(p^2 + q^2) \right)$$

$$\therefore x = a(p+q) \text{ i.e. } p+q = \frac{x}{a}$$

$$y = \frac{a}{2}(p^2 + q^2)$$

$$= \frac{a}{2}((p+q)^2 - 2pq)$$

$$\therefore y = \frac{a}{2}\left(\frac{x^2}{a^2} + 2\right) \text{ or } y = \frac{x^2}{2a} + a$$

since $pq = -1$ (a focal chord)

(c).

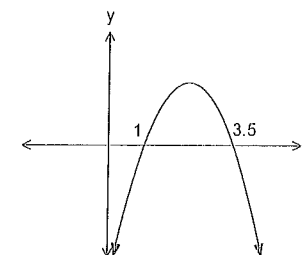
$$\frac{5}{x-1} \times (x-1)^2 > 2(x-1)^2$$

$$5(x-1) - 2(x-1)^2 > 0$$

$$(x-1)(5-2(x-1)) > 0$$

$$(x-1)(7-2x) > 0$$

$$1 < x < 3.5$$

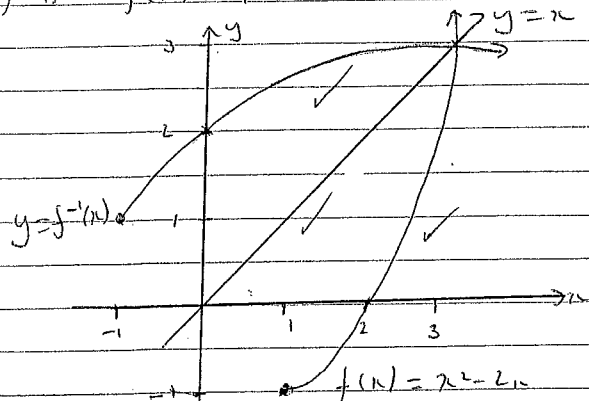


Extr 1

2009

Question 5.

a) i. $f(x) = x^2 - 2x$ for $x \geq 1$



-! if curves
do not meet
on $y=x$

-! if restriction
not apply.

(3)

ii) $x = y^2 - 2y$ ✓
 $= y^2 - 2y + 1 - 1$
 $= (y-1)^2 - 1$ (2)
 $(y-1)^2 = x+1$
 $y = 1 + \sqrt{x+1}$ ✓

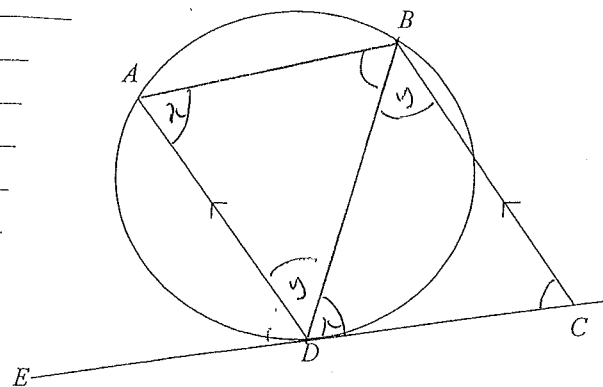
iii) $f^{-1}(2) = 1 + \sqrt{3}$ (1)
~~Not correct~~
 No curve through
 from trivial answer
 in ii)

b) $\frac{\pi}{2} = \frac{2\pi}{n}$, $a=3$ $v^2 = n^2(a^2 - x^2)$
 $n = \frac{2\pi}{\frac{\pi}{2}} = 4$ ✓ $v^2 = 16(9 - x^2)$
 $v = \sqrt{16(9 - x^2)}$
 $= 12 \text{ ms}^{-1}$ ✓

$a = -n^2 x$
 $= -16x$

max acceleration when $x = -3$
 $a = -16(-3)$
 $= 48 \text{ ms}^{-2}$ ✓

(3)



In Δ 's DBC, ABD

$\angle BDC = \angle BAD$ ($\angle$ in alt seg. ment) ✓

$\angle CBD = \angle BDA$ (alt $\angle$'s $AD \parallel BC$) ✓

$\therefore \Delta ABC \parallel \Delta DCB$ (equiangular)

$\frac{BD}{AD} = \frac{BC}{BD}$ (corres sides sim Δ 's)

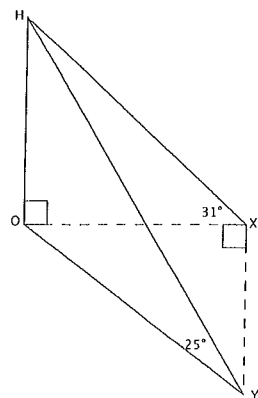
$BD^2 = AD \cdot BC$

$\sqrt{\frac{2}{1} \times \frac{2}{1}} = \frac{5}{6} \times \frac{10}{11} \times \frac{9}{7} \times \frac{5}{13}$
 select
 1/11

(3)

Question 6:

a)



In $\triangle OHX$:

$$\tan 31^\circ = \frac{h}{OX}$$

$$OX = \frac{h}{\tan 31^\circ}$$

In $\triangle OHY$:

$$\tan 25^\circ = \frac{h}{OY}$$

$$OY = \frac{h}{\tan 25^\circ}$$

In $\triangle OXY$:

$$OY^2 = OX^2 + XY^2$$

$$\frac{h^2}{\tan^2 25^\circ} = \frac{h^2}{\tan^2 31^\circ} + 150^2$$

$$\frac{h^2}{\tan^2 25^\circ} - \frac{h^2}{\tan^2 31^\circ} = 150^2$$

$$\frac{h^2 \tan^2 31^\circ - h^2 \tan^2 25^\circ}{\tan^2 25^\circ \tan^2 31^\circ} = 150^2$$

$$h^2 (\tan^2 31^\circ - \tan^2 25^\circ) = 150^2 \tan^2 25^\circ \tan^2 31^\circ$$

$$h^2 = \frac{150^2 \tan^2 25^\circ \tan^2 31^\circ}{\tan^2 31^\circ - \tan^2 25^\circ}$$

$$h = 110.91096 \dots$$

$$h \approx 110.9 \text{ m (to 1 dec. pl)}$$

b) $\sqrt{3} \cos \theta - \sin \theta = R \cos(\theta + \alpha)$

$$= R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = 1$$

$$R = \sqrt{(\sqrt{3})^2 + 1^2}$$

$$= 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$2 \cos \left(\theta + \frac{\pi}{6} \right) = 1$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = \frac{1}{2}$$

$$\theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\theta = \frac{\pi}{6}, \frac{3\pi}{2}$$

c) (i)

$$y = \frac{8}{x^2 + 4}$$

$$= 8(x^2 + 4)^{-1}$$

$$y' = \frac{-16x}{(x^2 + 4)^2}$$

At $x = 2$:

$$m_T = \frac{-16 \times 2}{(2^2 + 4)^2}$$

$$= -\frac{32}{64}$$

$$= -\frac{1}{2}$$

For normal: $m_N = -\frac{1}{m_T}$
 $= 2$

At $x = 2$:

$$y = \frac{8}{2^2 + 4}$$

$$= 1$$

Equation of normal at $(2, 1)$:

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 2)$$

$$y = 2x - 3$$

(ii) Points of intersection:

$$2x - 3 = \frac{8}{x^2 + 4}$$

$$(2x - 3)(x^2 + 4) = 8$$

$$2x^3 + 8x - 3x^2 - 20 = 0$$

$$\text{Let } P(x) = 2x^3 + 8x - 3x^2 - 20$$

$$P(1) = -13$$

$$P(-1) = -33$$

$$P(2) = 0$$

$$\therefore (x - 2) \text{ is a factor of } P(x)$$

Using division of polynomials:

$$P(x) = (x - 2)(2x^2 + x + 10)$$

$$\Delta = 1^2 - 4 \cdot 2 \cdot 10$$

$$= -79$$

As $\Delta < 0$, $2x^2 + x + 10 = 0$ has no real roots. (i.e. no more points of intersection).

So the normal $y = 2x - 3$ does not touch the curve again.

7(a) $f(x+3)$ is $f(x)$ translated

3 units to the left

$$\int_{-4}^7 f(x+3) dx = \int_{-1}^7 f(x) dx \checkmark$$

$$= \int_{-1}^0 f(x) dx + \int_0^7 f(x) dx$$

$$A = \int_0^1 f(x) dx + \int_0^7 f(x) dx$$

$$= 1^2 \times 2^1 + 7^2 \times 2^7$$

$$= 6274 \checkmark$$

(i)(ii) $\ddot{x} = 0$ $\dot{y} = -g$

$$\dot{x} = V \cos \alpha \checkmark \quad \dot{y} = V \sin \alpha - gt$$

$$x = Vt \cos \alpha \quad y = Vt \sin \alpha - \frac{1}{2}gt^2$$

(iii) Let $y = 0$

$$t(V \sin \alpha - \frac{1}{2}gt) = 0 \checkmark$$

$$V \sin \alpha = \frac{1}{2}gt$$

$$\frac{2V \sin \alpha}{g} = t \checkmark$$

(iii) $\tan \beta = \frac{dy}{dx} \checkmark$

$$= \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \frac{V \sin \alpha - gt}{V \cos \alpha}$$

$$V \cos \alpha \tan \beta = V \sin \alpha - gt$$

$$gt = V \sin \alpha - V \cos \alpha \tan \beta$$

$$= \frac{V \sin \alpha \cos \beta - V \cos \alpha \sin \beta}{\cos \beta}$$

$$t = \frac{V \sin(\alpha - \beta)}{g \cos \beta} \checkmark$$

(iv) $\frac{2V \sin \alpha}{3g} = \frac{V \sin(\alpha - \frac{\alpha}{2})}{g \cos \frac{\alpha}{2}} \checkmark$

$$\frac{2 \sin \alpha}{3g} = \frac{\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}}$$

$$\frac{2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}}{3} = \frac{\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}}$$

$$\frac{4 \cos^2 \frac{\alpha}{2}}{3} = 1 \quad \text{since } \sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 \frac{\alpha}{2} = \frac{3}{4} \checkmark$$

$$\cos \frac{\alpha}{2} = \frac{\sqrt{3}}{2}$$

$$\frac{\alpha}{2} = 30^\circ$$

$$\alpha = 60^\circ \checkmark$$